

Thermal Noise (Resistor Noise, Johnson-Nyquist Noise)

→ Random Electron Movement

→ freq characteristic is white



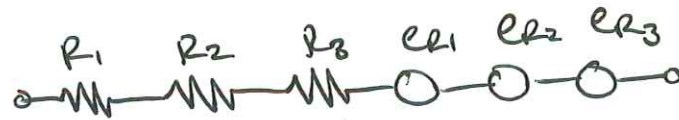
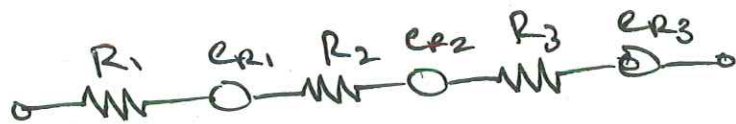
$$e_R^2 = 4kTR$$

$$e_R = \sqrt{4kTR}$$

$k \rightarrow$ Boltzmann's constant
 $1.38 \times 10^{-23} \text{ J/K}$

$T \rightarrow$ Temperature (in K)

Resistors in series:



$$R_{eq} = R_1 + R_2 + R_3$$

$$e_{Reg}^2 = e_{R1}^2 + e_{R2}^2 + e_{R3}^2$$

$$e_{Reg}^2 = 4kTR_1 + 4kTR_2 + 4kTR_3 = 4kT(R_1 + R_2 + R_3) = 4kTR_{eq}$$

Total noise only dependent on R_{eq} , not individual resistors
(this holds true for parallel)

flicker Noise

→ Noise generated by DC movement of charge (DC current)
specifically through passives.

→ $1/f$ type noise

$$\bar{i}_n^2 = \frac{k I_{DC}^a}{f}$$

k → NOT Boltzmann's constant!

a, k just constants, defined device-to-device.

Both in range $[0.5, 2]$

Rule of thumb: → Resistors that wire-wound → very low flicker
bulk silicon → high flicker

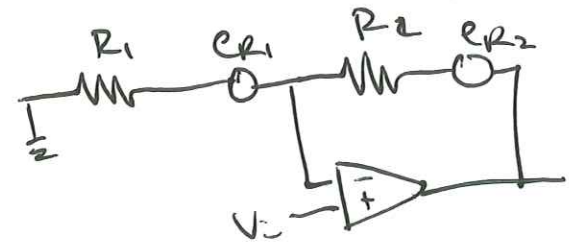
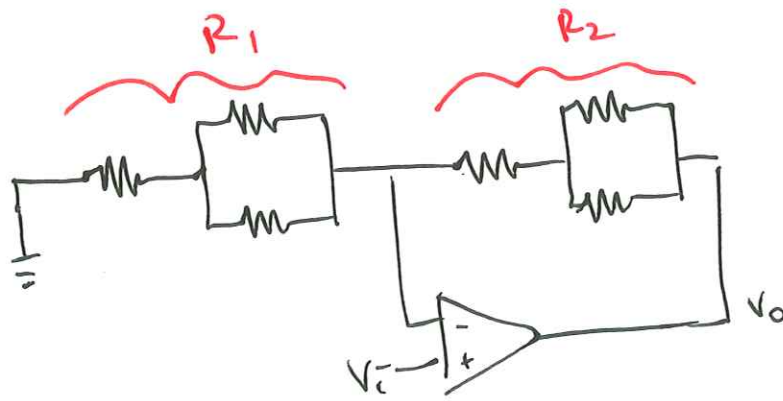
Shot Noise: Noise produced by charges passing through barriers (eg. P-N junction)

$$\bar{i}_n^2 = 2q I_{DC}$$

← Not dependent on f ! → white noise

q → electron charge = $1.602 \times 10^{-19} \text{C}$

Ex:

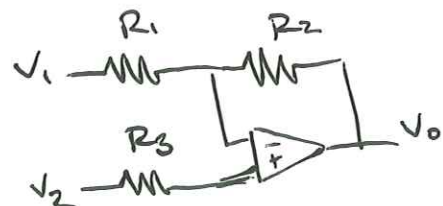


Resistor noise instead as a current!

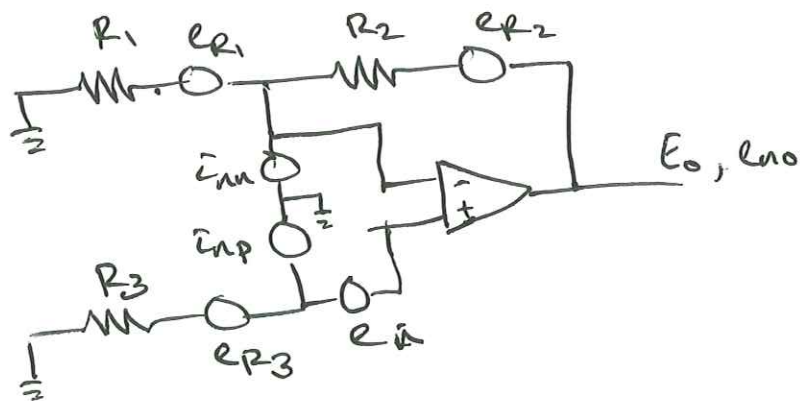


$$\bar{i}_{Rn}^2 = \frac{e_R^2}{R^2} = \frac{4kT}{R} \rightarrow \bar{i}_{Rn} = \sqrt{\frac{4kT}{R}}$$

Noise from a Generic Amp

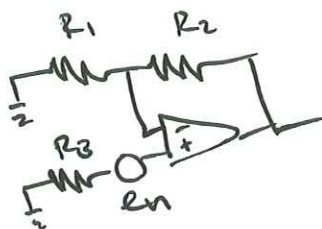


noise



Superposition:

1) $e_n \neq 0$

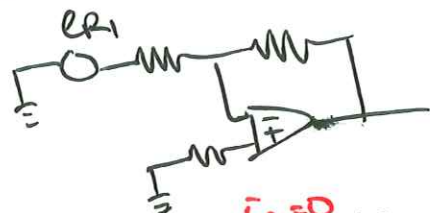


$$e_{no,1} = e_n \left(1 + \frac{R_2}{R_1}\right)$$

2) $e_{R3} \neq 0 \rightarrow$ same as e_n

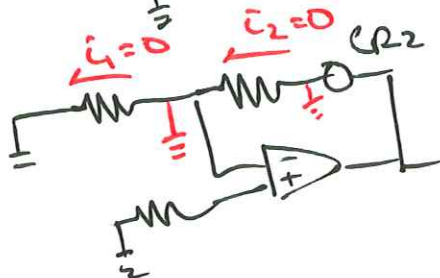
$$e_{no,2} = e_{R3} \left(1 + \frac{R_2}{R_1}\right)$$

3) $e_{R1} \neq 0$



$$e_{no,3} = e_{R1} \frac{R_2}{R_1}$$

4) $e_{R2} \neq 0$

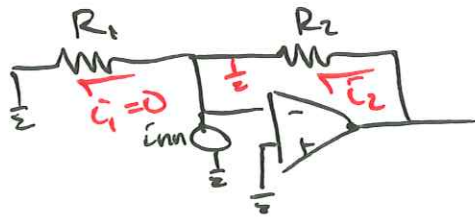


$$e_{no,4} = e_{R2}$$

(gain technically $-\frac{R_2}{R_1}$ but - implies phasing)

$\rightarrow$ continued.

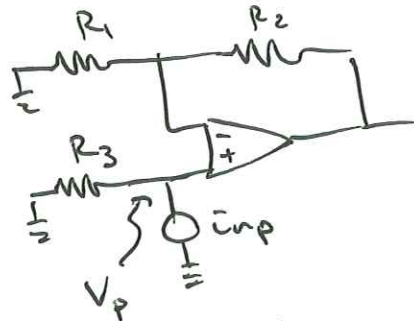
5) $\bar{i}_{nn} \neq 0$:



$$\bar{i}_2 = \bar{i}_{nn}$$

$$e_{no,5} = \bar{i}_{nn} R_2$$

6) $\bar{i}_{np} \neq 0$:



$$V_p = \bar{i}_{np} R_3$$

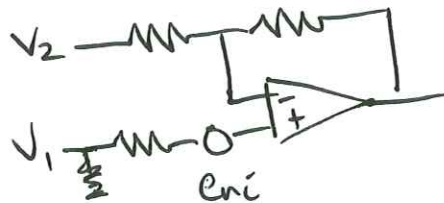
$$e_{no,6} = \left(1 + \frac{R_2}{R_1}\right) \bar{i}_{np} R_3$$

to combine:

$$e_{no}^2 = \sum_{i=0}^6 e_{no,i}^2 = \left(\cancel{e_n^2 + e_{R3}^2 + \bar{i}_{np}^2} \right) \left(1 + \frac{R_2}{R_1}\right)^2 + e_{R1}^2 \frac{R_2^2}{R_1^2} + e_{R2}^2 + \bar{i}_{nn}^2 R_2^2$$

... could be simplified some...

Can define a total noise input a e_{ni} :



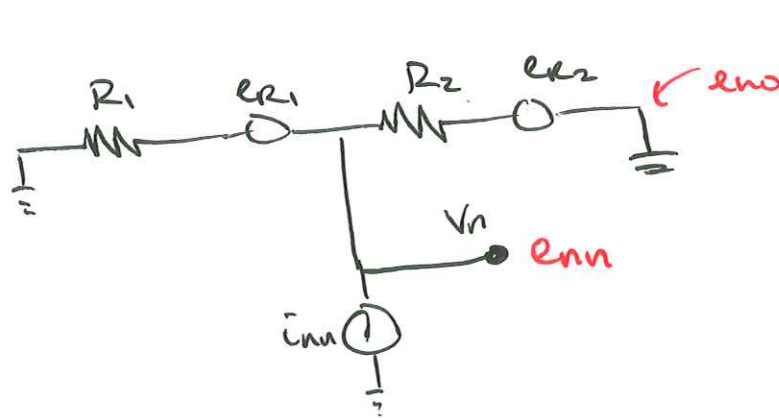
$$e_{no} = |A_n| e_{ni} \rightarrow e_{ni} = e_{no} / |A_n|$$

for this case, $A_n \neq A$

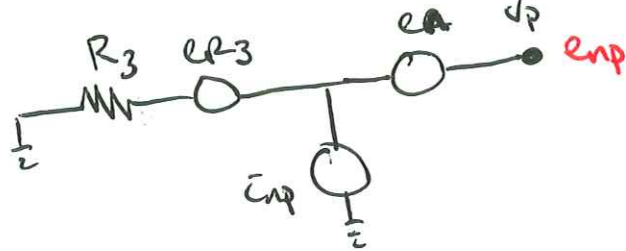
$$A_n = 1 + \frac{R_2}{R_1}$$

Calculate e_{ni} directly;

→ Assuming $V_o, V_i = 0$ ($e_{no} = 0$)



$$e_{nn}^2 = e_{n1}^2 \left(\frac{R_2}{R_1 + R_2} \right)^2 + e_{n2}^2 \left(\frac{R_1}{R_1 + R_2} \right)^2 + i_{nn}^2 R_2^2$$



$$e_{np}^2 = e_{n3}^2 + e_{n4}^2 + i_{np}^2 R_3^2$$

↓

$$e_{ni}^2 = e_{nn}^2 + e_{np}^2$$

$$e_{no}^2 = |A_n|^2 e_{ni}^2$$

Actual output of Amplifier:

$$V_o = V_1 \left(-\frac{R_2}{R_1} \right) + V_2 \left(1 + \frac{R_2}{R_1} \right) \pm e_{no}$$

↓ really should be $\pm e_{no}$

Bandwidth of amplifier effects on noise output.

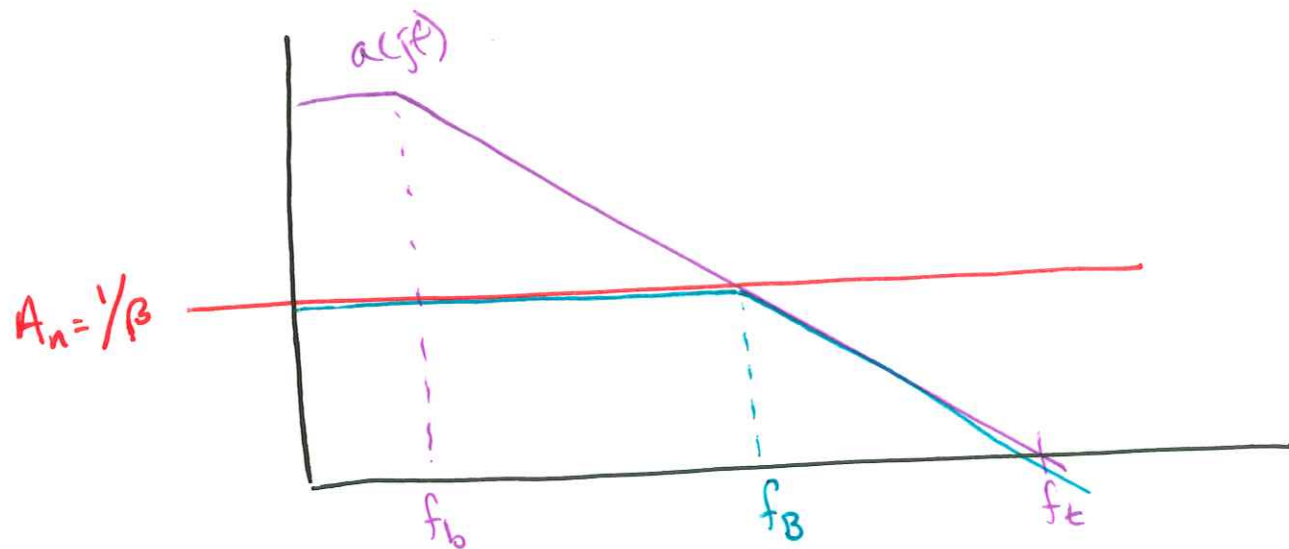
→ Previous solution assumed $a \rightarrow \infty$ for all f .

Solution 1) Calculate A_n as frequency dependent.

2) Use ideal A_n then add f_B after the fact.

Note $A_n = 1/\beta$ (non-inverting gain)

for example:



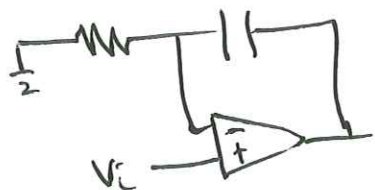
$$\rightarrow A_n(jf) = A_n \left(\frac{1}{1 + jf/f_B} \right)$$

$$f_B = \beta f_t$$

$$e_{no}(jf) = |A_n(jf)| e_{ni}(f) = |A_n| \left| \frac{1}{1 + jf/f_B} \right| e_{ni}(f)$$

Example

Modified Book 7.24



Non-inv. Integrator

$$R = 159 \text{ k}\Omega$$

$$C = 1 \text{ nF}$$

with low ESR

$$T = 20^\circ\text{C}$$

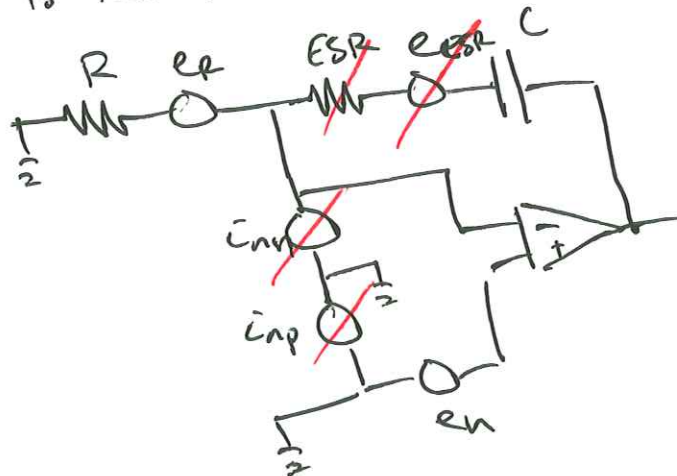
$$\text{Amp: } e_{nw} = 18 \text{ nV}/\sqrt{\text{Hz}}$$

$$f_{ce} = 200 \text{ Hz}$$

$$e_{nw} = 0$$

$$f_t = 3 \text{ MHz}$$

What is the total noise output power above 1 Hz?



$$\bar{e}_{nn}^2 = \bar{e}_{np}^2 = \bar{e}_{nw}^2 \left(\frac{f_{ce}}{f} + 1 \right)$$

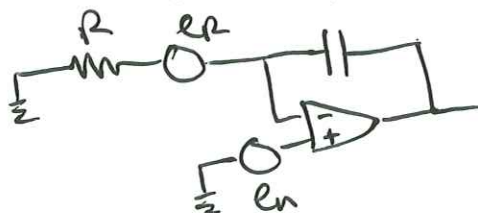
$$\bar{e}_{nn} = \bar{e}_{np} = 0$$

$$e_n^2 = e_{nw}^2 \left(\frac{f_{ce}}{f} + 1 \right)$$

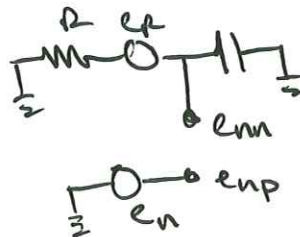
since ESR is low (on order of $< 1 \Omega$,
then $R \gg ESR$, ~~assumed~~

then $e_R \gg e_{ESR}$, $e_{ESR} \rightarrow 0$

Also say $ESR \rightarrow 0$



enc:



$$e_{nn}^2 = e_R^2 \left(\frac{Z_C}{R + Z_C} \right)^2 = e_R^2 \left(\frac{1/j\omega C}{R + 1/j\omega C} \right)^2$$

$$= e_R^2 / (1 + j\omega RC)^2$$

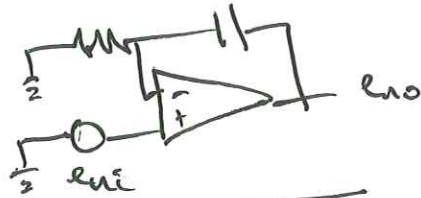
$$e_{np}^2 = e_n^2$$

$$= e_R^2 / |1 + j\omega RC|^2$$

$$= e_R^2 / (1 + \omega^2 R^2 C^2)$$

$$e_{nc}^2 = e_n^2 + e_R^2 / (1 + \omega^2 R^2 C^2)$$

Noise gain:



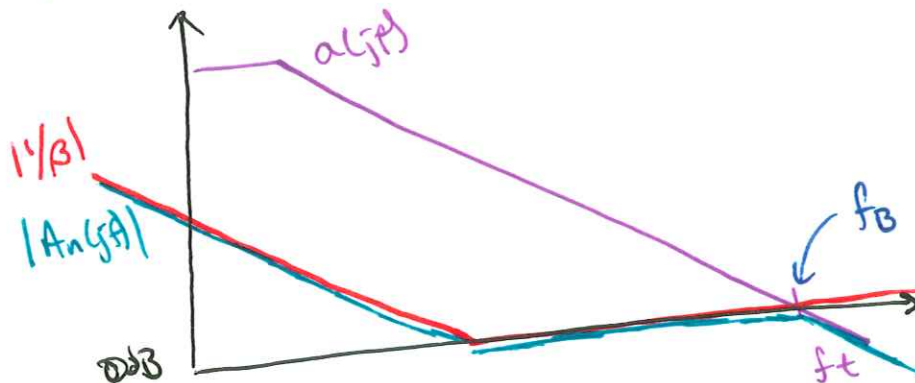
$$e_{ni} = e_{no} \frac{R}{R + j\omega C} \rightarrow \frac{e_{no}}{e_{ni}} = \frac{R + j\omega C}{R} = \frac{1 + j\omega RC}{j\omega RC}$$

$$A_n = \frac{1}{\beta} = \frac{1 + j\omega RC}{j\omega RC}$$

want $|A_n| = \frac{\sqrt{1 + (\omega RC)^2}}{\omega RC}$

$$e_{no}^2 = |A_n|^2 e_{ni}^2 = \frac{1 + (\omega RC)^2}{(\omega RC)^2} \left[e_n^2 + \frac{e_p^2}{1 + (\omega RC)^2} \right] = \underbrace{e_n^2 \left(1 + \frac{1}{(\omega RC)^2} \right)}_{\star} + \frac{e_p^2}{(\omega RC)^2}$$

Again, A_n doesn't consider $a(jf)$ effects... add them



$$A_n(jf) = A_n \left(\frac{1}{1 + jf/f_c} \right)$$

want $|A_n(jf)|^2 = |A_n|^2 \left| \frac{1}{1 + jf/f_c} \right|^2$

$$|A_n(jf)|^2 = \frac{1 + (\omega RC)^2}{(\omega RC)^2} \frac{1}{1 + f^2/f_c^2}$$

$$e_{no}^2 = \frac{1}{1 + f^2/f_c^2} \left[\star \right]$$

$$e_n^2 = e_{nw}^2 \left(\frac{f_{ce}}{f} + 1 \right) \quad e_p^2 = 4kTR$$

$$E_{no}^2 = \int_{1\text{Hz}}^{\infty} e_{no}^2(f) df$$

→ numerically solved in Matlab!

$$E_{no}^2 = 37 \text{ nV}^2 \rightarrow E_{no} = 192 \text{ nV}$$

Part 2: what is the SNR for input signal @ 100Hz with 1mV Amplitude.
(assume sine)

$$SNR = 20 \log_{10} \left(\frac{V_{sRMS}}{E_{no}} \right) = 10 \log_{10} \left(\frac{V_{sRMS}^2}{E_{no}^2} \right)$$

$$V_{iRMS} = 1mV / \sqrt{2} = 0.707 \text{ mV}$$

$$V_{ORMS} = V_{sRMS} = |A(f=100)| V_{iRMS} = 7.07 \text{ mV}$$

$$SNR = 20 \log_{10} \left(\frac{7.07 \text{ mV}}{192 \text{ mV}} \right) = 31.3 \text{ dB}$$